Task 1

```sql
SELECT *
FROM flights
JOIN airline
    ON flights.airline_id = airline.airline_id
WHERE airline.airline_name = 'YLQ';
```

Query | Query History

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	flight_id integer	flight_no character varying (50)	scheduled_departure date	scheduled_arrival date	departure_airport_id integer	arrival_airport_id integer	departing_gate character varying (50)	arriving_gate character varying (50)	airline_id integer	statu chara
1	85	US-IL	2023-08-27	2023-08-07	4	12	598	3445	5	Boar
2	174	PK-BA	2023-12-15	2023-10-04	17	16	3200	2618	5	Boar
3	251	MX-BCN	2024-03-07	2023-09-22	7	7	489	20	5	Dela
4	311	KI-G	2023-07-10	2023-09-02	8	20	10	1493	5	Boar
5	391	MH-NMU	2024-03-01	2023-06-05	7	4	23	39	5	Boar
6	662	CR-SJ	2023-05-20	2023-05-19	17	12	3287	74	5	Boar
7	823	US-FL	2023-08-06	2023-09-21	10	20	10	88	5	Boar
8	913	BR-MS	2023-06-21	2023-11-16	11	20	1568	9	5	Dela
9	916	JP-20	2023-03-25	2024-02-21	4	10	2182	256	5	Boar
10	920	KR-11	2023-05-15	2023-06-23	16	13	59	244	5	Dela

Explanation

This query displays all flights operated by a specific airline. The JOIN clause connects the flights table with the airline table using the common airline_id column. The WHERE clause filters the results and shows only flights operated by the airline named 'YLQ'.

Task 2

```sql
SELECT f.flight_id, f.flight_no,
       ap.airport_name AS departure_airport
FROM flights f
JOIN airport ap
    ON f.departure_airport_id = ap.airport_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 1

	flight_id integer	flight_no character varying (50)	departure_airport character varying (50)
1	1	US-CT	Elorza Airport
2	2	US-NM	Figari Sud-Corse Airport
3	3	FI-OL	Darchula Airport
4	4	RU-KR	Lime Acres Finsch Mine Airport
5	5	RO-DJ	Hana Airport
6	6	CA-SK	Darchula Airport
7	7	AU-TAS	Ocean Falls Seaplane Base
8	8	US-AZ	Figari Sud-Corse Airport

Explanation

The flights table is joined with the airport table using the departure_airport_id and airport_id columns. The aliases f and ap simplify table references. The AS keyword renames the airport_name column to departure_airport in the result.

Task 3

```sql
SELECT airline.*
FROM airline
LEFT JOIN flights
    ON airline.airline_id = flights.airline_id
    AND flights.scheduled_departure >= CURRENT_DATE + INTERVAL '1 month'
    AND flights.scheduled_departure < CURRENT_DATE + INTERVAL '2 months'
WHERE flights.flight_id IS NULL;
```

Data Output | Messages | Notifications

Showing rows: 1 to 51 | Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
1	1	SCIP	IPC	Russia	2023-12-02	2023-09-18
2	2	YPDA	PDN	Brazil	2024-01-15	2023-11-23
3	3	FKKH	KLE	Slovenia	2023-12-16	2023-05-23
4	4	OOKB	KHS	France	2023-11-18	2024-02-18
5	5	CYLQ	YLQ	Kazakhstan	2023-03-23	2023-07-26
6	6	FANG	NGL	Panama	2023-11-27	2024-02-03
7	7	SSZS	0	Iran	2023-11-02	2024-01-16
8	8	SNIG	QIG	Greece	2023-12-07	2023-08-01
9	9	KNQX	NQX	Venezuela	2024-03-16	2023-09-16
10	10	LFKS	SOZ	Indonesia	2023-07-10	2023-09-20

Total rows: 51 | Query complete 00:00:00.142 | CRLF | Ln 13, Col 18

Explanation

This query finds airlines with no flights scheduled for next month. The LEFT JOIN and IS NULL condition identify airlines without flights during the specified period.

Task 4

```sql
SELECT p.*
FROM passengers p
JOIN booking b
    ON p.passenger_id = b.passenger_id
JOIN booking_flight bf
    ON b.booking_id = bf.booking_id
JOIN flights f
    ON bf.flight_id = f.flight_id
WHERE f.flight_no = 'US-LA';
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

	passenger_id [PK] integer	first_name character varying (50)	last_name character varying (50)	date_of_birth date	gender character varying (50)	country_of_citizenship character varying (50)	country_of_residence character varying (50)	passport_number character varying (50)	created_at date
1	166	Lazarus	Emmitt	1991-07-02	Male	China	Vietnam	431745091-7	2023-06-18
2	195	Aurthur	Ramel	1984-09-06	Male	China	China	756903529-1	2023-09-26
3	91	Nadya	Jaqueminet	2001-09-22	Female	Venezuela	Argentina	662917991-9	2024-02-02

Explanation

This query displays passengers booked on a specific flight. The JOIN clauses connect passengers, bookings, and flights using their corresponding IDs.

Task 5

```
SELECT flights.flight_id, flights.flight_no,
       AVG(booking.price) AS avg_price,
       SUM(booking.price) AS total_price,
       MAX(booking.price) AS max_price,
       MIN(booking.price) AS min_price
FROM flights
LEFT JOIN booking_flight
    ON flights.flight_id = booking_flight.flight_id
LEFT JOIN booking
    ON booking_flight.booking_id = booking.booking_id
GROUP BY flights.flight_id, flights.flight_no;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 1

	flight_id [PK] integer	flight_no character varying (50)	avg_price numeric	total_price numeric	max_price numeric	min_price numeric
1	652	BR-MA	[null]	[null]	[null]	[null]
2	273	AU-WA	8929.2000000000000000	8929.20	8929.20	8929.20
3	51	US-DE	[null]	[null]	[null]	[null]
4	951	PG-MPL	3034.5400000000000000	6069.08	5888.12	180.96
5	839	FR-U	[null]	[null]	[null]	[null]
6	70	MY-12	683.2700000000000000	683.27	683.27	683.27
7	350	CN-41	5936.7680000000000000	29683.84	9792.28	2520.04

Explanation

This query calculates the average, total, maximum, and minimum ticket prices for each flight. The LEFT JOIN includes flights even if they have no bookings.

Task 6

```
SELECT flights.*, airline.airline_name,
       airport.country AS destination_country
FROM flights
JOIN airport
    ON flights.arrival_airport_id = airport.airport_id
JOIN airline
    ON flights.airline_id = airline.airline_id
WHERE airport.country = 'Kazakhstan';
```

Data Output | Messages | Notifications

flight_id integer	flight_no character varying (50)	scheduled_departure date	scheduled_arrival date	departure_airport_id integer	arrival_airport_id integer	departing_gate character varying (50)	arriving_gate character varying (50)	airline_id integer	statu chara

Explanation

This query displays flights arriving in Kazakhstan, along with airline names and destination countries.

Task 7

```sql
SELECT p.passenger_id, p.first_name, p.last_name, p.date_of_birth, ap.airport_name AS arrival_airport
FROM passengers p JOIN booking b ON p.passenger_id = b.passenger_id JOIN booking_flight bf ON b.booking_id = bf.booking_id
JOIN flights f ON bf.flight_id = f.flight_id JOIN airport ap ON f.arrival_airport_id = ap.airport_id
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years';
```

Data Output | Messages | Notifications

Showing rows: 1 to 22 | Page No: 1 of 1

	passenger_id integer	first_name character varying (50)	last_name character varying (50)	date_of_birth date	arrival_airport character varying (50)
1	159	Vivyan	Mallabone	2009-11-01	Alert Bay Airport
2	159	Vivyan	Mallabone	2009-11-01	Hana Airport
3	159	Vivyan	Mallabone	2009-11-01	Elorza Airport
4	159	Vivyan	Mallabone	2009-11-01	Darchula Airport
5	159	Vivyan	Mallabone	2009-11-01	Darchula Airport
6	67	Bernie	Michal	2009-11-03	Zephyrhills Municipal Airport
7	184	Ronnica	Laybourn	2009-03-06	Garbaharey Airport
8	159	Vivyan	Mallabone	2009-11-01	Zephyrhills Municipal Airport

Explanation

This query displays passengers under 18 years old and their arrival airports. The JOIN clauses connect passengers with their booked flights and destinations.

Task 8

```sql
SELECT passengers.first_name || ' ' || passengers.last_name AS full_name,
       passengers.passport_number,
       flights.actual_arrival AS arrival_time
FROM passengers
JOIN booking
    ON passengers.passenger_id = booking.passenger_id
JOIN booking_flight
    ON booking.booking_id = booking_flight.booking_id
JOIN flights
    ON booking_flight.flight_id = flights.flight_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 1

	full_name text	passport_number character varying (50)	arrival_time date
1	Muhammad Fass	109932770-9	2023-07-18
2	Trevar Broun	788025864-7	2024-02-11
3	Auria Breffit	570537341-4	2023-07-11
4	Archie Toffel	677556708-1	2023-06-17
5	Sanders Biddles	514546405-3	2023-09-05
6	Sanders Biddles	514546405-3	2024-03-01
7	Remington Piggot	470074456-1	2023-05-31

Explanation

This query displays passengers' full names, passport numbers, and actual arrival times.

Task 9

```sql
SELECT airport.country AS origin_country,
       COUNT(flights.flight_id) AS flight_count
FROM flights
JOIN airline
    ON flights.airline_id = airline.airline_id
JOIN airport
    ON flights.departure_airport_id = airport.airport_id
WHERE airline.airline_country = airport.country
GROUP BY airport.country;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	origin_country character varying (50)	flight_count bigint
1	Indonesia	6
2	Slovenia	2
3	Greece	1
4	Russia	5
5	China	50
6	Brazil	5
7	Poland	2
8	Philippines	3

Explanation

This query counts flights where the airline's home country matches the departure airport's country, grouped by country.

Task 10

```sql
SELECT flights.flight_no, airline.airline_name, departure.airport_name AS departure_airport, arrival.airport_name AS arrival_airport
FROM flights
JOIN airline
    ON flights.airline_id = airline.airline_id
JOIN airport AS departure
    ON flights.departure_airport_id = departure.airport_id
JOIN airport AS arrival
    ON flights.arrival_airport_id = arrival.airport_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 1

	flight_no character varying (50)	airline_name character varying (50)	departure_airport character varying (50)	arrival_airport character varying (50)
1	US-CT	GYA	Elorza Airport	Ocean Falls Seaplane Base
2	US-NM	SMM	Figari Sud-Corse Airport	Zephyrhills Municipal Airport
3	FI-OL	SMM	Darchula Airport	Elorza Airport
4	RU-KR	BLD	Lime Acres Finsch Mine Airport	Longana Airport
5	RO-DJ	PIR	Hana Airport	Alert Bay Airport
6	CA-SK	SMM	Darchula Airport	Akunaq Heliport
7	AU-TAS	SOZ	Ocean Falls Seaplane Base	Darchula Airport
8	US-AZ	KOQ	Figari Sud-Corse Airport	Hana Airport
9	IN-OR	IFH	Hana Airport	Pitalito Airport

Explanation

This query displays flight numbers, airline names, departure airports, and arrival airports.

Task 11

```
SELECT passengers.first_name || ' ' || passengers.last_name AS full_name, flights.flight_no
FROM passengers
JOIN booking
    ON passengers.passenger_id = booking.passenger_id
JOIN booking_flight
    ON booking.booking_id = booking_flight.booking_id
JOIN flights
    ON booking_flight.flight_id = flights.flight_id
JOIN airline
    ON flights.airline_id = airline.airline_id
WHERE airline.airline_name = 'YLQ';
```

Data Output | Messages | Notifications

Showing rows: 1 to 7 | Page No: 1 of 1

	full_name text	flight_no character varying (50)
1	Fred Simmgen	JP-20
2	Anetta Frank...	JP-20
3	Vanda Prime	JP-20
4	Margarita As...	KI-G
5	Joanie Oaks	KI-G
6	Nathanil July...	CR-SJ
7	Hermy Shearer	CR-SJ

Explanation

This query displays passengers who booked flights operated by YLQ, along with their full names and flight numbers.

Task 12

```
SELECT flights.flight_no, airport.country AS destination_country, COUNT(booking_flight.booking_id) AS passenger_count
FROM flights
JOIN airport
    ON flights.arrival_airport_id = airport.airport_id
LEFT JOIN booking_flight
    ON flights.flight_id = booking_flight.flight_id
GROUP BY flights.flight_id, flights.flight_no, airport.country;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 1

	flight_no character varying (50)	destination_country character varying (50)	passenger_count bigint
1	RU-ORL	Indonesia	1
2	RU-KR	China	0
3	US-WA	Poland	0
4	NA-OT	Greece	0
5	MY-14	Indonesia	2
6	CA-AB	Afghanistan	1
7	US-OK	China	0
8	PG-SAN	China	0
9	US-OH	Philippines	0

Explanation

This query counts passengers for each flight and displays the destination country, including flights with no passengers.

Task 13

```
95  SELECT passengers.*
96  FROM passengers
97  LEFT JOIN booking
98      ON passengers.passenger_id = booking.passenger_id
99  WHERE booking.booking_id IS NULL;
```

Data Output | Messages | Notifications

Showing rows: 1 to 16 | Page No: 1 of 1

	passenger_id [PK] integer	first_name character varying (50)	last_name character varying (50)	date_of_birth date	gender character varying (50)	country_of_citizenship character varying (50)	country_of_residence character varying (50)	passport_number character varying (50)	created_a date
1	82	Boote	Laurisch	1996-09-12	Male	Philippines	Bahamas	780593281-6	2023-09-(
2	31	Miquela	Tattersfield	1978-04-16	Female	Brazil	China	288290736-2	2023-09-'
3	146	Rudiger	Leith-Harvey	1976-05-18	Male	Canada	Philippines	630961668-4	2023-12-'
4	193	Sigfrid	Finding	2005-11-25	Male	Peru	Japan	989370779-X	2023-11-:
5	123	Conn	Philipps	1987-05-18	Male	Thailand	Indonesia	706168591-3	2023-08-:
6	96	Weylin	Dubery	1975-01-12	Male	China	Russia	711430427-7	2023-08-(
7	108	Fairfax	Brahmer	1991-02-10	Male	Morocco	Portugal	702800724-3	2023-08-(
8	24	Osborn	Golsby	1975-07-29	Male	Poland	Indonesia	063795036-4	2023-05-(

Explanation

This query displays passengers who have no bookings. The LEFT JOIN and IS NULL condition identify passengers without booked tickets.

Task 14

```
100 SELECT airline.airline_name, COUNT(flights.flight_id) AS total_flights
101 FROM airline
102 LEFT JOIN flights
103     ON airline.airline_id = flights.airline_id
104 GROUP BY airline.airline_id, airline.airline_name;
```

Data Output | Messages | Notifications

Showing rows: 1 to 51 | Page No: 1 of 1

	airline_name character varying (50)	total_flights bigint
1	SJS	14
2	NHT	29
3	KHS	13
4	SMM	19
5	Lab Five Airways	0
6	RBR	16
7	CII	24
8	YBQ	15
9	KMA	31

Explanation

This query counts the number of flights for each airline, including airlines with no flights.

Task 15

```sql
SELECT airline.airline_name, flights.flight_no, airport.city AS departure_city
FROM flights
JOIN airline
    ON flights.airline_id = airline.airline_id
JOIN airport
    ON flights.departure_airport_id = airport.airport_id
WHERE airport.country = 'Kazakhstan';
```

Data Output | Messages | Notifications

airline_name	flight_no	departure_city
character varying (50)	character varying (50)	character varying (50)

Explanation

This query displays airlines, flight numbers, and departure cities for flights departing from airports in Kazakhstan.